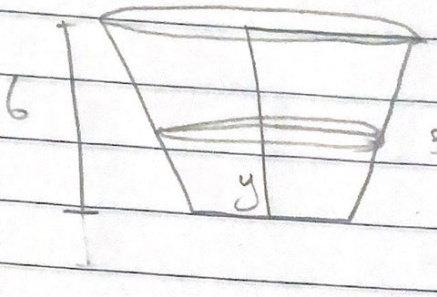
Kanageshwaran Dhakshinamoorthi
Parker Cruce
Miles Noland

Quiz 7:

CS:.

①



$$V = \pi\left(3+\frac{1}{3}y\right)^2 \Delta y$$

$$W = \int_0^6 (62.4)\pi\,(6-y)\left(3+\frac{1}{3}y\right)^2 dy$$

$$W = (62.4\pi)\int_0^6 (6-y)\left(9+\frac{6}{3}y-\frac{y^2}{9}\right) dy$$

$$W = 62.4\pi \int_0^6 \left(54+3y-\frac{4y^2}{3}-\frac{y^3}{9}\right) dy$$

$$W = 62.4\pi \left[54y+\frac{3y^2}{2}-\frac{4}{9}y^3-\frac{y^4}{36}\right]_0^6$$

$$W = (62.4\pi)(324+54-96-36)-(0)$$

$$W = 15350.4\pi$$

$$\boxed{W = 48224.70387\ J}$$